
EE 616: Microwave engineering

Course Code	EE 616
Course Name	Microwave engineering
L-T-P-C	2-0-0-2
Prerequisites	Basic Course in Electromagnetics / Engineering Electromagnetics
Intended for	Senior B.Tech, Mtech, PhD students of Electrical Engineering
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Transmission Line Theory and network analysis:** Electromagnetic analysis and transmission line theory of coaxial lines and waveguides, metamaterial lines, Impedance matching in microwave circuits, Microwave network analysis, N-port microwave network, Scattering matrix, Properties of the scattering matrix, S-parameters at arbitrary planes, S parameter measurements. [5 Lectures]
- **Power Dividers and Directional Couplers:** Basic Properties of power dividers and couplers, T junction, Wilkinson type, quadrature hybrid power dividers, coupled line directional coupler, 180 deg. Hybrid coupler. [4 Lectures]
- **Microwave Filters:** Basic Filter design techniques like image parameter and insertion loss, Filter transformations and implementations, low pass filters, coupled line filters, coupled resonator based filters, metamaterial filters. [4 Lectures]
- **Ferrite devices:** Circulators, isolators, phase shifters. [04 hrs]
- **Active Microwave components:** RF Diode, Microwave Transistors, Microwave ICs. [4 Lectures]
- **Microwave sources and Amplifiers:** Tube type sources like magnetrons, klystrons, Single stage Transistor amplifier, Stability circles, Broad band amplifier design, Solid state Power amplifiers. [4 Lectures]
- **Oscillators and Mixers:** Microwave oscillators using Transistors, dielectric resonators, diode and transistor based mixers. [3 Lectures]

Text Books

1. David. M. Pozar, **Microwave Engineering**, Wiley.

References

1. R.E. Collins, **Foundations for Microwave Engineering**, IEEE Press.